

# k\_davey\_mom\_keltner - Strategy Walkthrough

Strategy name k\_davey\_mom\_keltner Files reviewed - src/strategies/k\_davey\_mom\_keltner/strategies/k\_davey\_mom\_keltner/backtest\_k\_davey\_mom\_keltner\_v2.py Step-by-step execution

Backtest fetches OHLCV and warns if timeframe is not 1d. 2) It computes keltner\_stoch, trend\_stops, and ATR volatility filter plus its rolling average. 3) Base equity and contract multiplier (special handling for MES symbol). 4) BacktesterV2 is initialized in futures pnl\_mode with ATR multipliers by side. 5) Main loop starts after the maximum warmup across momentum, Keltner, ATR, and volatility windows. 6) Pending actions are executed at next bar open (ENTRY or EXIT). 7) If no position exists and ATR is valid, stop is updated from average price via bt.update\_stop\_from\_avg. 8) bt.on\_bar checks intrabar stop execution. 9) Net equity and dynamic contract count are computed. 10) bt.trade\_size is updated each bar. 11) Entry conditions evaluate momentum, trend, volatility stochastic thresholds. 12) Exit conditions set next-bar EXIT when Keltner stochastic exceeds specific exit threshold. 12) Outputs are generated: stats, equity curve, optional report, and parameters.

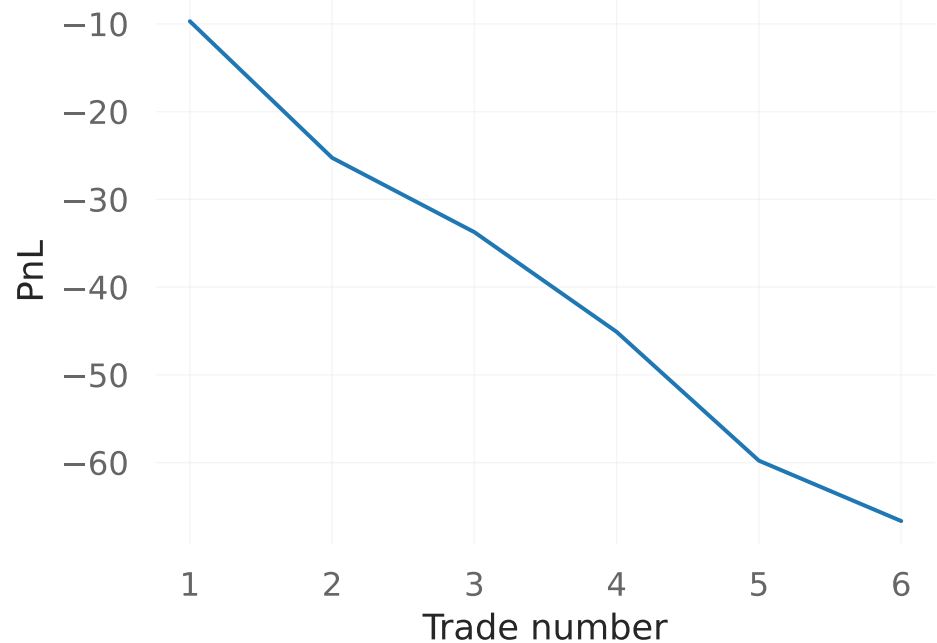
Parameters and defaults - Momentum and Keltner: mom\_length\_long=40, mom\_length\_short=40, keltner\_length=5, keltner\_atr\_mult=0.5. - Thresholds and trend: entry\_threshold=30.0, exit\_threshold=70.0, trend\_ema=200. - Volatility filter: volatility\_atr\_period=14, volatility\_sma\_period=100, volatility\_mult=0.8. - Position sizing: use\_position\_sizing=True, base\_equity=15000.0, sizing\_factor=0.33, max\_contracts=15. - Execution and costs: initial\_balance=15000.0, position\_mode="fixed", commission\_pct=0.001, slippage\_pct=0.001. - Direction and risk: allow\_trailing\_stop=True, atr\_period=14, atr\_sl\_mult\_long=4.0, atr\_sl\_mult\_short=1.0, pyramiding=1, position\_pct=0.0. - Indicators/calculations - Keltner stochastic proxy:  $100 * (\text{close} - \text{lower\_kc}) / (\text{upper\_kc} - \text{lower\_kc})$ . - Momentum checks: close vs close N bars ago (separate N for long/short). - trend\_ema filter from close. - ATR for stop updates and initial ATR-based risk. - Volatility filter:  $\text{ATR}(\text{volatility\_atr\_period}, \text{rolling\_mean\_ATR} * \text{volatility\_mult})$ . - Dynamic contracts from net\_equity/base\_equity formula (if position\_pct path). Entry conditions - LONG setup requires all: - price > close[mom\_length\_long bars ago] - price > trend\_ema - volatility filter true - keltner\_stoch < entry\_threshold - position, or existing LONG with room for pyramiding. - SHORT setup (if allow\_short) mirrors LONG: - price < close[mom\_length\_short bars ago] - price < trend\_ema - volatility filter true - keltner\_stoch > (100 - entry\_threshold) - no position, or existing SHORT with room for pyramiding. Entries are queued and executed on next bar open. Exit conditions - If LONG and keltner\_stoch > exit\_threshold, queue EXIT for next bar open. - If SHORT and keltner\_stoch < (100 - exit\_threshold), queue EXIT for next bar open. - Additional exits can happen intrabar from ATR stop handling in bt.on\_bar. BacktesterV2. Risk and position management (SL/TP/ATR/trailing/pyramiding/position sizing) - based stop distances with different multipliers for LONG and SHORT. - No fixed percent SL/TP. - Position runner (both set to None in BacktesterV2 init). - Dynamic position size (contracts) based on progression, capped by max\_contracts. - Optional position\_pct path overrides formula-based sizing. - Pyramiding supported and checked with len(bt.lots) < bt.pyramiding. - Futures PnL mode with commission multiplier (5 for MES, else 50 default path). Outputs (trades, stats, plots, reports) - clean equity curve via bt.stats(). - Equity curve PNG under static/backtests/k\_davey\_mom\_keltner/<run\_id>/. - Trades and stats CSV under static/backtests/k\_davey\_mom\_keltner/<run\_id>/. - Extensive strategy metadata. - QuantStats report when enabled. Known ambiguities or inconsistencies in the code - Function signature includes generate\_plots but no trade-plot generation is implemented. - base\_equity default in signature is 15000.0, but fallback block uses 1500.0 for MES only when position\_pct is None. - Parameters named ema\_fast/ema\_slow appear in CSV schema although this strategy does not use EMA crossover periods.

## k\_davey\_mom\_keltner - Trade summary

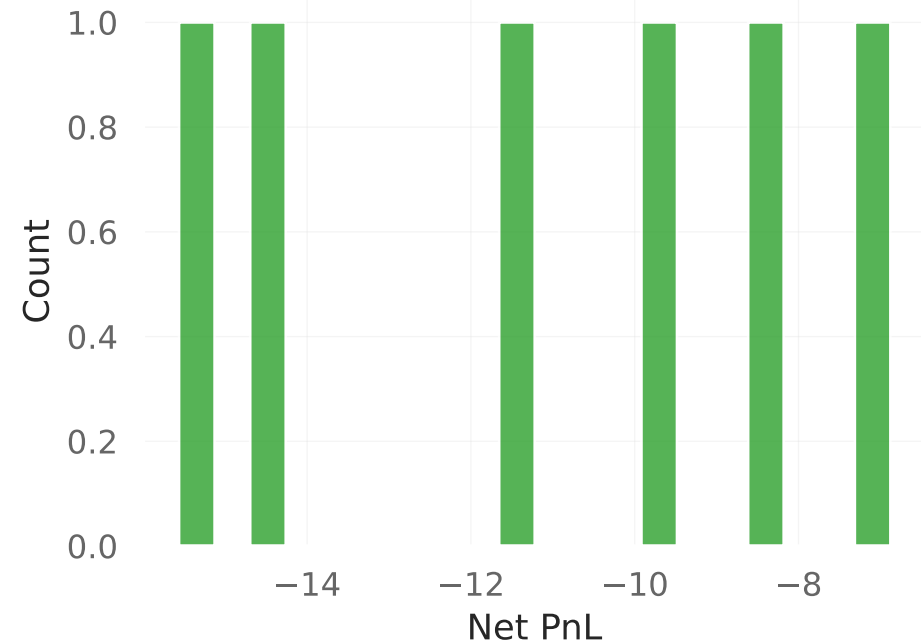
Trades: 6 Total Net PnL: -66.65 Average Net PnL per trade: -11.11 Win rate: 0.00% First trigger: Momentum+Keltner long Last trigger: Keltner stoch exit Source CSV: docs/strategy\_walkthroughs/csv\_baselines/k\_davey\_mom\_keltner.csv

# k\_davey\_mom\_keltner - Trade analytics

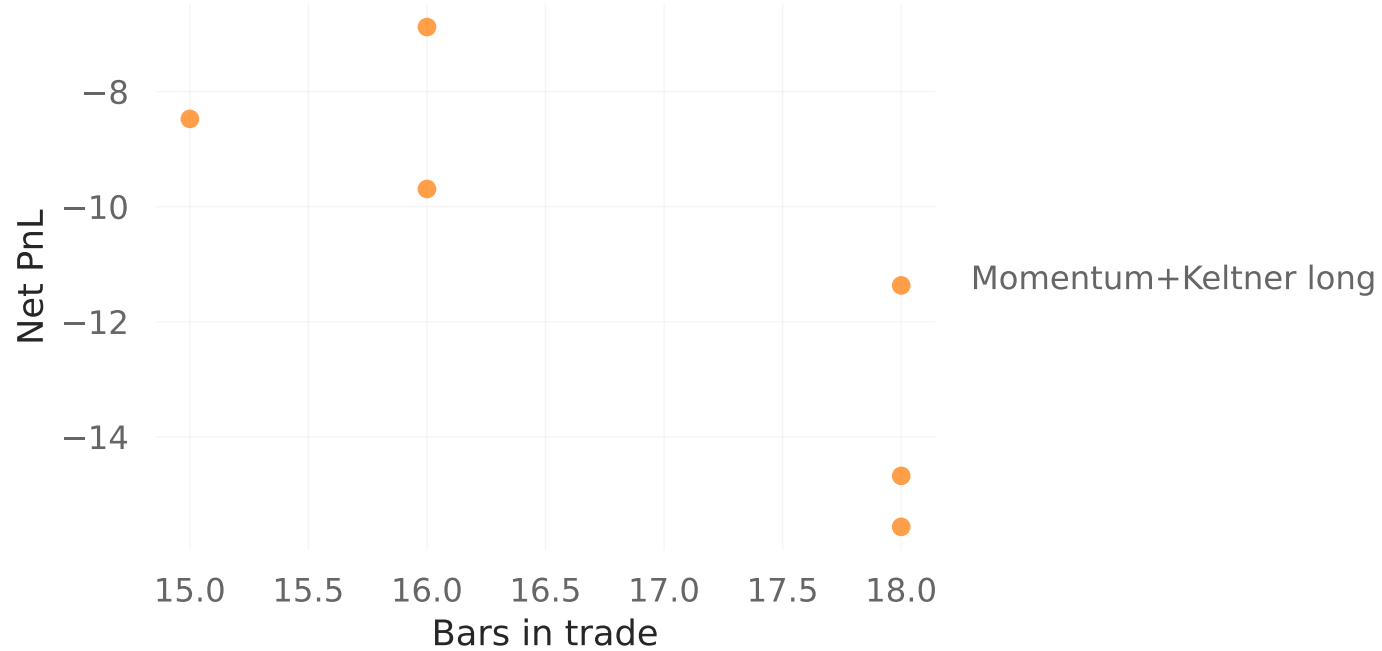
## Cumulative Net PnL by trade



## Net PnL distribution



## Duration vs Net PnL



## Top entry triggers

